

TECHNICAL USER GUIDE

Sword Mesh Trail System

Procedural, layered sword trails for Unity 6+

SYSTEM SUMMARY Generates a double-sided procedural ribbon between two tracked points, supports reusable presets, multiple material layers, Animation Events, Catmull–Rom smoothing, vertex-alpha fading, and per-renderer dissolve animation.

Hovl Studio

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IMPORTANT The component creates runtime MeshRenderer objects and a dynamic mesh automatically. Do not manually add a Trail Renderer component; this system builds its own geometry.

1. System Overview

HS_SwordMeshTrail generates a procedural ribbon mesh between Trail Point A and Trail Point B while the weapon moves. New trail sections are sampled in LateUpdate, old sections expire according to Trail Lifetime, and the mesh is rebuilt with continuous 0–1 UV coordinates.

The system is designed for sword slashes and other fast-moving melee effects, but it can also be used on staffs, claws, axes, energy blades, or any object where a ribbon should be generated between two tracked points.

Core features

- Automatic placement of Trail Point A and Trail Point B from local geometry bounds.
- Reusable ScriptableObject presets with one or more material layers.
- A separate Sorting Order, dissolve property, and dissolve target for every material layer.
- Square 0–1 UV layout with the newest section fixed at $U = 0$.
- Optional width subdivisions from 0 to 10 additional lines.
- Optional vertex-alpha fade controlled by an Animation Curve.
- Catmull–Rom interpolation for smoother curves when frame sampling is sparse.
- Automatic ParticleSystem control for effects below Trail Point A.
- Animation Event and public-method control.
- MaterialPropertyBlock-based dissolve values, allowing many slashes to share one material safely.

2. Included Scripts

Script	Purpose
HS_SwordMeshTrail	Main component. Samples the two points, generates the mesh, applies presets, controls material layers, dissolve, alpha, smoothing, and point effects.
HS_SwordTrailPreset	Reusable ScriptableObject containing trail geometry, smoothing, rendering, material-layer, dissolve, automatic-point, and effect-prefab settings.
HS_SwordTrailAnimationEvents	Receiver placed on the Animator object. Exposes StartSwordTrail, StopSwordTrail, and ClearSwordTrail to animation clips.

3. Quick Start

1. Select the sword or weapon object that contains the geometry used to determine the blade bounds.
2. Add the HS_SwordMeshTrail component.
3. Use Recalculate Automatic Trail Points from the component context menu. The component creates Trail Point A and Trail Point B if they do not exist.
4. Assign a Trail Material directly, or create and assign an HS_SwordTrailPreset.
5. Use Create/Connect Animation Events Component. The script finds the nearest Animator in the parent hierarchy and connects HS_SwordTrailAnimationEvents.
6. Add StartSwordTrail and StopSwordTrail Animation Events to the attack animation.
7. Enter Play Mode and confirm that the trail starts and stops at the intended frames.

RECOMMENDED WORKFLOW Use presets for production assets. Direct component fields are useful for a single trail, while presets make material layers and settings reusable across many slash prefabs.

Creating a preset

1. In the Project window, select Assets > Create > Hovl > Sword Trail Preset.
2. Configure Trail, Low FPS Curve Smoothing, Automatic Trail Points, Material Layers, Rendering, and Trail Point A Effects.
3. Assign the preset to HS_SwordMeshTrail.
4. Run Apply Trail Preset on the component to refresh copied values, layer objects, and effect-prefab instances.

4. Component Settings Reference

Preset

Preset. Optional HS_SwordTrailPreset asset. At startup, reusable values are copied into the component. Material-layer data remains available from the preset when runtime renderers are configured.

Trail

Setting	Default	Description
Trail Material	None	Fallback material used when no preset material layers are configured.
Dissolve Property Name	_Dissolve	Fallback shader Float reference name used when no preset material layers are configured.
Maximum Dissolve	0	Target dissolve value. A value of 0 disables script control completely.
Trail Lifetime	0.35 s	How long every sampled section remains alive after its spawn time.
Minimum Section Distance	0.015	Minimum movement of either point required before a new section is recorded.
Sample Interval	0	Minimum time between samples. Zero allows sampling every frame.
Lines Along Trail	2	Additional lines across the ribbon width. Valid range: 0–10.
Fade Alpha Over Lifetime	On	Enables vertex-alpha fading from newest to oldest sections.
Alpha Over Lifetime	1 → 0	Curve evaluated by normalized section age. The material must multiply by vertex color alpha.
Start Active	Off	Automatically calls StartTrail when the component becomes enabled in Play Mode.
Clear Previous Trail On Start	On	Removes remaining sections before a new trail begins.

Low FPS Curve Smoothing

Setting	Default	Description
Smooth Low FPS	On	Generates intermediate Catmull–Rom sections between recorded samples.
Maximum Smoothed Section Distance	0.08	Approximate maximum distance between generated curve sections. Smaller values create more geometry.
Max Intermediate Sections Per Frame	8	Safety limit for extra sections between two recorded samples. Valid range: 0–32.

Automatic trail points

Setting	Default	Description
Point A / Point B	Auto	References to the two transforms that define ribbon width.

Setting	Default	Description
Automatic Axis	Longest	Uses the largest local bounds dimension, or explicitly Local X, Y, or Z.
Recalculate Points On Awake	On	Repositions points from geometry bounds at runtime. Disable after manual placement.
Point A Inset	0	Moves Point A inward from its calculated end.
Point B Inset	0	Moves Point B inward from its calculated end.
Endpoint Padding	0	Extends both endpoints. Negative values shorten the line.

Rendering

Sorting Order. Fallback sorting order used for the single direct material. Preset layers use their own Sorting Order values.

Receive Shadows. Controls shadow reception on every generated layer renderer.

Shadow Casting Mode. Controls shadow casting for every generated layer. Off is recommended for most transparent VFX.

5. Preset Workflow

A preset centralizes settings that should be reused across multiple sword or slash prefabs. It also defines the number of runtime material layers and optional effect prefabs instantiated below Trail Point A.

Applying and refreshing

1. Assign the preset in the Preset field.
2. Choose Apply Trail Preset after editing reusable values or changing effect prefabs.
3. The action copies trail, smoothing, point, and rendering values into the component.
4. Runtime material-layer objects are rebuilt, and preset effect instances below Trail Point A are refreshed.
5. At Awake, the component also applies preset values automatically.

EDITING NOTE If you manually reposition Trail Point A or Trail Point B, disable Recalculate Points On Awake; otherwise the automatic bounds calculation will overwrite the manual positions.

6. Material Layers and Dissolve

Every preset Material Layer creates a separate runtime MeshRenderer that uses the same generated mesh. This allows multiple visual passes—such as a core, glow, distortion, or additive edge—with an independent material and Sorting Order.

Layer field	Behavior
Material	Material rendered by this layer.
Sorting Order	Renderer sorting order for this layer. The shader render queue and transparency settings must also support the intended order.
Dissolve Property Name	Exact shader Float reference name, for example <code>_Dissolve</code> . This is the Reference value, not only the Inspector display name.
Maximum Dissolve	Target value reached during the post-StopTrail fade. Zero disables all script writes for this property.

Dissolve lifecycle

1. When StartTrail is called, the layer reads the current dissolve value directly from its material.
2. While the trail is active, that material value is preserved as the starting dissolve value.
3. When StopTrail is called, dissolve progresses over Trail Lifetime.
4. The target is the greater of the material's starting value and Maximum Dissolve, so the script never lowers dissolve below the value already set in the material.
5. When Maximum Dissolve is 0, the script does not read, reset, or overwrite the shader property.

Material start	Maximum Dissolve	Result after StopTrail
0.20	0.80	Interpolates from 0.20 to 0.80.
0.80	0.50	Stays at 0.80; it never decreases.
Any value	0	No script control; the material property is untouched.

Shared-material safety

Dissolve is applied with MaterialPropertyBlock on each generated MeshRenderer. The shared Material asset is not modified and no material instance is created. Multiple slashes can therefore use the same material while maintaining independent dissolve values and independent StartTrail/StopTrail timing.

AVOID Do not use `sharedMaterial.SetFloat` from another runtime script unless you intentionally want to change the material's starting dissolve value for every object that shares it.

7. UV Layout and Shader Requirements

UV0 mapping

Coordinate	Meaning
U = 0	Newest section beside the weapon; always the left side of the texture.
U = 1	Oldest visible end of the complete trail.
V = 0	Trail Point A edge.
V = 1	Trail Point B edge.
Range	The complete visible trail uses one continuous 0–1 × 0–1 UV square.

Lines Along Trail adds geometry subdivisions across V without splitting or repeating the UV layout. The mesh contains separate front and back vertices with opposite normals so lit materials can render both sides correctly.

Shader checklist

- Use UV0 to sample the trail texture.
- Use vertex color alpha if Fade Alpha Over Lifetime is enabled.
- Expose the dissolve control as a shader Float property and enter its exact Reference name in the layer.
- Configure transparent blending, render queue, depth write, and culling according to the intended VFX style.
- If the shader already performs two-sided rendering, remember that the generated mesh is also double-sided.

8. Low-FPS Curve Smoothing

The system records only real sampled weapon poses. During mesh rebuilding, optional Catmull–Rom interpolation evaluates intermediate points from neighboring samples. This produces a curved average direction instead of connecting sparse samples with straight midpoints.

Quality and cost

Setting	Change	Effect
Maximum Smoothed Section Distance	Lower	Smoother curve, more intermediate sections, higher CPU and vertex cost.
Maximum Smoothed Section Distance	Higher	Fewer sections, lower cost, more visible faceting.
Max Intermediate Sections Per Frame	Higher	Allows difficult fast curves to receive more subdivision.
Minimum Section Distance	Lower	Records more real samples during small movements.
Sample Interval	Higher	Reduces sampling frequency but can require stronger smoothing.
Lines Along Trail	Higher	Adds width geometry; useful for vertex deformation but increases vertices and triangles.

Suggested starting points

Target	Suggested approach
Desktop / high quality	Smooth Low FPS on; Maximum Smoothed Section Distance around 0.04–0.08; intermediate cap 8–16.
Mobile / many simultaneous trails	Increase the smoothed distance, lower the intermediate cap, and keep Lines Along Trail only as high as the shader needs.
Very fast weapons	Keep Sample Interval at 0, reduce Minimum Section Distance, and use smoothing to round sparse frame-to-frame motion.

PROFILE IN CONTEXT Actual cost depends on movement speed, frame rate, Trail Lifetime, number of material layers, and the number of simultaneous trail components.

9. Trail Point A Effects

A preset can contain effect prefabs that are instantiated as children of Trail Point A when the preset is applied. ParticleSystem references do not need to be assigned manually.

- StartTrail refreshes all ParticleSystem components below Trail Point A, enables inactive effect roots when needed, clears previous particles, and calls Play.
- StopTrail calls Stop with StopEmitting, allowing existing particles to finish naturally according to their own lifetimes.
- The search includes inactive child objects.
- Manually added ParticleSystem objects below Trail Point A are also detected.

10. Animation Events and Runtime API

Animation Event receiver

Animation Event	Action
StartSwordTrail	Calls HS_SwordMeshTrail.StartTrail().
StopSwordTrail	Calls HS_SwordMeshTrail.StopTrail().
ClearSwordTrail	Calls HS_SwordMeshTrail.ClearTrail() and removes the complete mesh immediately.

Public methods

Method	Behavior
StartTrail()	Starts emission, optionally clears the old trail, captures material dissolve start values, plays Point A particles, and records the first section.
StopTrail()	Stops new sections and begins post-stop dissolve. Existing sections expire over Trail Lifetime.
ClearTrail()	Immediately removes all recorded sections and clears the generated mesh.
ApplyTrailPreset()	Copies preset values, recalculates points when enabled, refreshes effect prefabs, and rebuilds runtime layers.

Method	Behavior
RecalculateTrailPoints()	Calculates Point A and Point B from geometry bounds and the selected axis.
EnsureTrailPoints()	Creates missing Trail Point A and Trail Point B children and renames legacy HS Trail Point names.
EnsureAnimationEventsComponent()	Finds the nearest parent Animator, adds the receiver if necessary, and connects this trail.

Read-only properties

PointA / PointB. Returns the active point transforms.

IsEmitting. Returns true while new trail sections are being generated.

Preset. Returns the currently assigned HS_SwordTrailPreset.

Runtime example

```
using UnityEngine;

public class WeaponAttack : MonoBehaviour
{
    [SerializeField] private Hovl.HS_SwordMeshTrail swordTrail;

    public void BeginAttack() => swordTrail.StartTrail();
    public void EndAttack() => swordTrail.StopTrail();
    public void CancelAttack() => swordTrail.ClearTrail();
}
```

11. Performance Guidelines

- Every active HS_SwordMeshTrail owns one dynamic generated mesh.
- Every material layer creates one runtime MeshRenderer and generally one draw call.
- All layers of one trail share the same generated mesh, avoiding duplicated geometry generation per material.
- Dissolve uses MaterialPropertyBlock, so it does not instantiate or modify shared materials.

- The main CPU cost is section sampling, Catmull–Rom subdivision, list population, and mesh rebuilding in LateUpdate.
- Long Trail Lifetime values keep more sections alive and increase mesh size.
- The system switches to 32-bit mesh indices only when the generated vertex count exceeds 65,535.
- Shadows, expensive transparent shaders, distortion, and multiple layers may cost more than the trail-generation script itself.

Optimization priority

1. Reduce the number of material layers.
2. Increase Maximum Smoothed Section Distance or reduce Max Intermediate Sections Per Frame.
3. Reduce Trail Lifetime.
4. Increase Minimum Section Distance.
5. Use Sample Interval only when less frequent sampling is acceptable.
6. Reduce Lines Along Trail when the shader does not need width subdivisions.
7. Disable shadow casting and receiving for transparent VFX unless visually required.

12. Troubleshooting

Symptom	Resolution
No trail is visible	Confirm that a material is assigned, StartTrail is called, the weapon moves enough to create at least two sections, and both trail points exist.
No material assigned warning	Assign Trail Material or add at least one valid Material Layer to the preset.
Trail width or position is wrong	Recalculate points, choose the correct axis, then adjust Point A Inset, Point B Inset, or Endpoint Padding.
Manual points move at runtime	Disable Recalculate Points On Awake after manual placement.
Trail looks angular	Enable smoothing, lower Maximum Smoothed Section Distance, increase the intermediate cap, or improve sampling frequency.
Texture direction is reversed	The newest weapon-side section is intentionally U = 0. Flip the texture or shader sampling if the artwork expects the opposite direction.
Alpha fade does not work	Enable Fade Alpha Over Lifetime and make the shader use vertex color alpha.
Alpha fade is unwanted	Disable Fade Alpha Over Lifetime. Vertex alpha will remain 1.

Symptom	Resolution
Dissolve does not change	Maximum Dissolve must be above 0, the Reference name must match exactly, and the material shader must expose that Float property.
Dissolve starts above zero	This is intentional: every trail reads the current property value from its material at StartTrail.
Dissolve does not move downward	This is intentional: the target cannot be lower than the material's starting value.
Layer order appears wrong	Check the layer Sorting Order, shader render queue, blend mode, and depth-write settings.
Point A particles do not play	Keep the prefab or ParticleSystem below Trail Point A and ensure it is valid and enabled by the component.
Animator warning	Place an Animator on the weapon's parent hierarchy or connect HS_SwordTrailAnimationEvents manually.

Render-pipeline compatibility

The C# mesh-generation system is not tied to a specific render pipeline. The assigned materials and shaders must be compatible with the project's Built-in, URP, or HDRP configuration. Transparent sorting, depth, lighting, and distortion behavior are shader- and pipeline-dependent.

13. Final Setup Checklist

- HS_SwordMeshTrail is attached to the intended weapon object.
- Trail Point A and Trail Point B are positioned correctly.
- Recalculate Points On Awake is disabled if the points were positioned manually.
- A direct material or preset Material Layer is assigned.
- The shader reads UV0 and, when needed, vertex color alpha.
- Every dissolve Reference name matches the shader property exactly.
- Maximum Dissolve is 0 for layers that must not be controlled by the script.
- Apply Trail Preset was run after preset or effect-prefab changes.
- HS_SwordTrailAnimationEvents is connected to the correct trail.
- StartSwordTrail and StopSwordTrail events are placed at the intended animation frames.
- The trail has been profiled with the expected number of simultaneous slashes.

READY The trail system is configured when the geometry follows the blade correctly, material layers render in the intended order, StopTrail produces the expected fade/dissolve, and repeated attacks reset cleanly.